Flex Optimization Problems

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1 Shortage Optimization

1.1 Notation

The notation for the shortage optimization problem is summarized in Table 1.

Table 1: Notation for the shortage optimization problem

Indices and sets	
$(i \in)\ I$	Items
$I^{FG} \subset I$	Finished goods
$I^{SFG} \subset I$	Semi-Finished Goods or Subassemblies
$I^{RM} \subset I$	Raw materials
$(t \in)\ T := \{1, \ldots, n_T\}$	Planning time horizon (days)
$(w \in)\ W := \{1, \ldots, n_W\}$	Demand "weeks" or intervals between two consecutive dates in ordered set of demand due dates. Demand due dates typically occur on specified days of the calendar week.
$(t^*) \in \{1, \ldots, n_{T,\mathcal{C}}\}$	Index set of working days calendar $\mathcal{C}$ for production or procurement of item i
$(t^{\dagger}_{t^*}) \in \mathcal{C}_{i \in I}$	Working days calendar for production or procurement of item i
$(g_j \in)\{G_j\}_{j \in I}$	BOM substitution groups index set. Group g_j is indexed by BOM parent item j
$\tilde{S}_{g_j} \subset I$	Children components within substitution group g_j
$(q_j \in)\{Q_j\}_{j \in I}$	Work order (WO) index set. WO q_j is indexed by WO supply item j
Parameters	
$\beta_{i,j,t} \geq 0$	BOM structures: if production starts at time $t \in T$, then $\beta_{i,j,t}$ units of item $i \in I$ are used to produce one unit of item $j \in I$ (default: 0)
$\tilde{\beta}_{i,g_j,t} \geq 0$	BOM substitution structures: if production starts at time $t \in T$, then $\tilde{\beta}_{i,g_j,t}$ units of item $i \in \tilde{S}_{g_j}$ are used to produce one unit of item $j \in I$ (default: 0)
$R_i \geq 0$	Per-unit revenue (in \$) for item $i \in I$ (default: 0)
$\gamma_{g_j,i} \geq 0$	Cost for using BOM subsititution component $i \in \tilde{S}_{g_j}$ to make item j
$S_{i,t} \geq 0$	Supply (in units) of item $i \in I$ scheduled to arrive at time $t \in T$ (default: 0, pre-expediting)
$D_{i,w} \geq 0$	Demand (in units) of item $i \in I$ and $w \in W$
$Inv_{i,0} \geq 0$	Initial inventory (in units) of item $i \in I$ (default: 0)
$LT_i \geq 0$	Lead time (in days) of item $i \in I$ (if $i \in I^{FG}$, then this is a manufacturing lead time; if $i \in I^{BW}$, then this is a supply lead time)
$\hat{LT}_i \geq 0$	Lead time (in days) of item $i \in I$ (if $i \in I^{FG}$, then this is a manufacturing lead time; if $i \in I^{BW}$, then this is a supply lead time)
$IRR \geq 0$	Internal Rate of Return (unitless): Flex's risk-free rate of return to incentivize having cash on hand (default: 10%)
$n^{Sho} \geq 0$	Number of resolved raw material shortages
$C_{i,t} \geq 0$	Per-unit cost for panic supply of item $i \in I$ at time $t \in T$ (default: 1)
$P_{i,t} \geq 0$	Upper bound on panic supply $p_{i,t}$
$\gamma_{g_j,i}$	Cost penalty for using component i within BOM substitution group g_j
$\mathcal{Q}_i^{max}$	Maximum order/production quantity for item $i \in I$
$\mathcal{Q}_i^{min}$	Minimum order/production quantity for item $i \in I$
$\mathcal{Q}_i^{mul}$	Multiple order/production quantity (rounding value) for item $i \in I$. Any produced/ordered quantity can only be in integer multiples of this parameter
$\hat{S}_{q_j}$	Work order q_i's supply quantity on item j
$\hat{D}_{q_j,i}$	Dependent demand quantity on component i for work order q_j
$\hat{\Delta}_{q_j}$	Remaining days from work order due date to end date
$\hat{n_T}$	Horizon length between the earliest work order due date and optimization end date
c_1	Work order fulfillment discount factor parameter
ϵ	Small positive number
Variables	
Continuous	
$Inv_{i,t}$	Inventory of item $i \in I$ at time $t \in T$
$p_{i,t}$	Panic supply of item $i \in I$ at time $t \in T$ (to resolve raw material shortage)
$\delta_{i,t}$	Infinite supply beyond procurement lead time on item $i \in I^{RM}$ at time $t \in T$.
$Obj^{Rev} \geq 0$	Revenue objective
$Obj^{WO} \geq 0$	Work order (WO) fulfilment objective,
$Obj^{BOMsub} \geq 0$	BOM susbtituion priority objective,

Binary	
Sho_i	1 if shortage for raw material $i \in I^{BW}$ is resolved, 0 otherwise
$\mathbb{1}^y_{i,t}$	Production indicator variable, 1 if production of item $i \in I$ starts at time $t \in T$
$\mathbb{1}^p_{i,t}$	Panic supply indicator variable, 1 if panic supply of item $i \in I$ needed at time $t \in T$
$z_t^{\hat{S}_{q_j}}$	1 if production for WO $q \in Q$ and item $j \in I$ started at time $t \in T$
$\bar{z}_t^{\hat{S}_{q_j}}$	1 if production for WO $q \in Q$ and item $j \in I$ finished at time $t \in T$
Integer	
$y_{i,t}$	Daily Demand met for item $i \in I$ at time $t \in T$
$d_{i,w}$	Weekly demand met for item $i \in I$ at week $w \in W$
$x_{i,t}$	# of units of item $i \in I$ for which production starts at time $t \in T$
$\bar{x}_{i,t}$	# of units of item $i \in I$ for which production ends at time $t \in T$
$\tilde{x}_{g_j,i,t}$	# of units of assembly j of BOM substition group $g_j \in \{G_j\}_{j \in I}$ for which production starts at time $t \in T$ and uses substitution item $i \in \tilde{S}_{g_j}$
$z^y_{i,t}$	Rounding value multipliers for demand met $y_{i,t}$ to be in multiple quantities of $\mathcal{Q}_i^{mul}$
$z^p_{i,t}$	Rounding value multipliers for panic supply $p_{i,t}$ to be in multiple quantities of $\mathcal{Q}_i^{mul}$

1.2 Problem Formulation

We wish to solve the following Mixed-Integer Linear Program (MILP):

Objective:

We wish to maximize work order (WO) fulfilment, revenue generation and usage of prioritizied BOM substitution components (the α parameters can be used to weigh the objectives as desired):

$$\max_{\mathbf{y},\mathbf{d},\mathbf{x},\bar{\mathbf{x}},\tilde{\mathbf{x}},\mathbf{z}^{\hat{\mathbf{S}}},\bar{\mathbf{z}}^{\hat{\mathbf{S}}},\mathbf{z}^{\mathbf{y}},\mathbf{z}^{\mathbf{p}},\mathbf{p},\delta,\mathbb{1}^{\mathbf{p}},\mathbb{1}^{\mathbf{y}},Inv} \quad \alpha^{Rev}Obj^{Rev} + \alpha^{WO}Obj^{WO} - \alpha^{BOMsub}Obj^{BOMsub} \tag{1}$$

Constraints:

Calculate the revenue objective disincentivized for panic supply over utilization:

$$Obj^{Rev} = \sum_{i\in I}\sum_{t\in T}(1+IRR)^{\frac{n_T-t}{365.25}}R_i y_{i,t} - C_{i,t}p_{i,t}, \tag{2}$$

Calculate the WO fulfilment objective:

$$Obj^{WO} = \sum_{q_j\in\{Q_j\}_{j\in I}}\sum_{t\in T} c_1^{\frac{n_T-t}{n_T}}\left(\frac{\hat{\Delta}_{q_j}}{\hat{n_T}}+\epsilon\right)\bar{z}_t^{\hat{S}_{q_j}}, \tag{3}$$

Calculate the BOM substitution priority objective:

$$Obj^{BOMsub} = \sum_{g_j\in\{G_j\}_{j\in I}}\sum_{i\in\tilde{S}_{g_j}}\sum_{t\in T}\gamma_{g_j,i}\tilde{x}_{g_j,i,t}, \tag{4}$$

Mass balance:

$$\begin{aligned} Inv_{i,t} = Inv_{i,t-1} + S_{i,t} + \sum_{q_j\in\{Q_j\}_{j\in I}:j=i}\bar{z}_t^{\hat{S}_{q_j}}\hat{S}_{q_j} + p_{i,t} + \bar{x}_{i,t} + \delta_{i,t} - \\ y_{i,t} - \sum_{j\in I:\ \beta_{i,j,t}>0}\beta_{i,j,t}x_{j,t} - \sum_{g_j\in\{G_j\}_{j\in I}:\tilde{\beta}_{i,g_j,t}>0}\tilde{\beta}_{i,g_j,t}\tilde{x}_{g_j,i,t} - \\ \sum_{q_j\in\{Q_j\}_{j\in I}:\hat{D}_{q_j,i}>0} z_t^{\hat{S}_{q_j}}\hat{D}_{q_j,i} \qquad i\in I,\ t\in T, \end{aligned} \tag{5}$$

Non-working day:

$$
\begin{aligned}
x_{i,t} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && i \in I, t \in T,\\
\bar{x}_{i,t} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && i \in I, t \in T,\\
p_{i,t} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && i \in I, t \in T,\\
y_{i,t} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && i \in I, t \in T,\\
z_t^{\hat{S}_{q_j}} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && q_j \in \{Q_j\}_{j \in I}, t \in T,\\
\bar{z}_t^{\hat{S}_{q_j}} &= 0 \quad \text{if } t \notin \mathcal{C}_i, && q_j \in \{Q_j\}_{j \in I}, t \in T,
\end{aligned}
\tag{6}
$$

Bounding x (we cannot produce raw materials):

$$
0 \leq x_{i,t^\dagger_{t^*}} \leq \begin{cases} +\infty & \text{if } i \notin I^{RM} \text{ and } t^* \leq n_{T,\mathcal{C}_i} - LT_i, \\ 0 & \text{if } i \in I^{RM} \text{ or } t^* > n_{T,\mathcal{C}_i} - LT_i, \end{cases}, \qquad i \in I,\ t^\dagger_{t^*} \in \mathcal{C}_i,
\tag{7}
$$

Bounding $\bar{x}$ (we cannot produce raw materials):

$$
0 \leq \bar{x}_{i,t^\dagger_{t^*}} \leq \begin{cases} +\infty & \text{if } i \notin I^{RM} \text{ and } t^* > LT_i, \\ 0 & \text{if } i \in I^{RM} \text{ or } t^* \leq LT_i, \end{cases}, \qquad i \in I,\ t^\dagger_{t^*} \in \mathcal{C}_i,
\tag{8}
$$

Bounding y:

if demand can be met beyond due dates, bound across horizon:

$$
0 \leq y_{i,t} \leq \sum_{w \in W} D_{i,w}, \qquad i \in I, t \in T
$$

if demand cannot be met beyond due dates, bound across current and future demand weeks:

$$
0 \leq y_{i,t} \leq \sum_{w' \geq w} D_{i,w'}, \qquad i \in I, t \in T, t \twoheadrightarrow w
\tag{9}
$$

Bounding d:

if demand can be met beyond due dates, bound across horizon

$$
\sum_{w \in W} d_{i,w} \leq \sum_{w \in W} D_{i,w}, \qquad i \in I, w \in W
$$

if demand cannot be met beyond due dates

$$
\begin{aligned}
0 \leq d_{i,w} &\leq \sum_{w' \geq w} D_{i,w'}, && i \in I, w \in W\\
\sum_{w' \geq w} d_{i,w'} &\leq \sum_{w' \geq w} D_{i,w'}, && i \in I, w \in W
\end{aligned}
\tag{10}
$$

Link daily demand met to weekly demand met:

$$
d_{i,w} = \sum_{t \in T : t \twoheadrightarrow w} y_{i,t} \qquad i \in I, w \in W
\tag{11}
$$

Link starting and finishing production via the manufacturing lead time:

$$x_{i,t^\dagger_{t^*}} = \bar{x}_{i,t^\dagger_{t^{*'}}}, \qquad i \in I,\ t^\dagger_{t^*} \in \mathcal{C}_i :\ t^* \leq n_{T,\mathcal{C}_i} - LT_i,\ t^{*'} = t^* + LT_i, \tag{12}$$

Rounding values (min/max/multiple quantity) constraint on demand met

$$\begin{aligned} y_{i,t} &\geq \mathbb{1}^y_{i,t} \mathcal{Q}^{min}_i, && i \in I, t \in T \\ y_{i,t} &\leq \mathbb{1}^y_{i,t} \mathcal{Q}^{max}_i, && i \in I, t \in T \\ y_{i,t} &= z^y_{i,t} \mathcal{Q}^{mul}_i, && i \in I, t \in T \end{aligned} \tag{13}$$

BOM substitution: bounding components usage:

$$\sum_{i \in \tilde{S}_{g_j}} \tilde{x}_{g_j,i,t} = x_{j,t}, \qquad g_j \in \{G_j\}_{j \in I}, t \in T \tag{14}$$

Work orders: bounding $z^{\hat{S}}$:

$$\begin{aligned} 0 \leq z^{\hat{S}_{q_j}}_{t^\dagger_{t^*}} &\leq \begin{cases} 1 & \text{if } t^* \leq n_{T,\mathcal{C}_i} - \hat{LT}_i, \\ 0 & \text{otherwise} \end{cases}, && q_j \in \{Q_j\}_{j \in I},\ t^\dagger_{t^*} \in \mathcal{C}_i, \\ \sum_{t \in T} z^{\hat{S}_{q_j}}_t &\leq 1, && q_j \in \{Q_j\}_{j \in I} \end{aligned} \tag{15}$$

Work orders: bounding $\bar{z}^{\hat{S}}$:

$$\begin{aligned} 0 \leq \bar{z}^{\hat{S}_{q_j}}_{t^\dagger_{t^*}} &\leq \begin{cases} 1 & \text{if } t^* > \hat{LT}_i, \\ 0 & \text{otherwise} \end{cases}, && q_j \in \{Q_j\}_{j \in I},\ t^\dagger_{t^*} \in \mathcal{C}_i, \\ \sum_{t \in T} \bar{z}^{\hat{S}_{q_j}}_t &\leq 1, && q_j \in \{Q_j\}_{j \in I} \end{aligned} \tag{16}$$

Work orders: link starting and finishing production via the dependent demand lead time:

$$z^{\hat{S}_{q_j}}_{t^\dagger_{t^*}} = \bar{z}^{\hat{S}_{q_j}}_{t^\dagger_{t^{*'}}}, \qquad q_j \in \{Q_j\}_{j \in I},\ t^\dagger_{t^*} \in \mathcal{C}_i :\ t^* \leq n_{T,\mathcal{C}_i} - \hat{LT}_i,\ t^{*'} = t^* + \hat{LT}_i, \tag{17}$$

Infinite raw material supply slack beyond lead time:

$$0 \leq \delta_{i,t^\dagger_{t^*}} \leq \begin{cases} +\infty & \text{if } i \in I^{RM} \text{ and } t^* > n_{T,\mathcal{C}_i}, \\ 0 & \text{otherwise}, \end{cases} \qquad i \in I,\ t^\dagger_{t^*} \in \mathcal{C}_i \tag{18}$$

(Deprecated) Panic supplies to resolve raw material shortages:

$$0 \leq p_{i,t} \leq \begin{cases} M\ Sho_i & \text{if } i \in I^{BW,todo} \text{ and } t > LT_i, \\ P_{i,t} & \text{if } i \in I^{BW,done} \text{ and } t > LT_i, \\ 0 & \text{otherwise}, \end{cases} \qquad i \in I,\ t \in T, \tag{19}$$

(Deprecated) We bound the number of resolved raw material shortages:

$$\sum_{i \in I} Sho_i \leq n^{Sho}, \tag{20}$$

2 Excess & Obsolete Inventory Optimization

2.1 Notation

The notation for the excess & obsolete (E&O) inventory optimization problem is summarized in Table 2.

Table 2: Notation for the E&O inventory optimization problem.

Indices and sets	
$(i, j \in) I$	Items
$I^{BW} \subset I$	Raw materials (items ending in "-BW")
$I^{FG} \subset I$	Finished goods (items ending in "-FG")
Parameters	
$\beta_{i,j} \geq 0$	BOM structures: $\beta_{i,j}$ units of raw material $i \in I^{BW}$ are used to produce one unit of finished good $j \in I^{FG}$ (if not given, then default: 0)
$S_i \geq 0$	Maximum available supply (in units) of raw material $i \in I^{BW}$ (default: ∞)
$D_i \geq 0$	Maximum available demand (in units) of finished good $i \in I^{FG}$ (default: ∞)
$Inv_{i,0} \geq 0$	Initial inventory (in units) of raw material $i \in I^{BW}$ (if not given, then default: 0)
$C_i \geq 0$	Per-unit cost of sourcing raw material $i \in I^{BW}$ (if not given, then default: 0)
$B \geq 0$	Maximum raw material investment (budget, must be given as main hyper-parameter)
Variables	
Continuous	
$Inv_{i,1}$	Inventory of raw material $i \in I^{BW}$ after production/depletion
x_i	Units of raw material $i \in I^{BW}$ sourced
y_i	Units of finished good $i \in I^{FG}$ produced

2.2 Problem Formulation #1

For this variation of the problem, we define B (likle in Table 2) as the maximum raw material investment (budget). This problem solves the following:

What is the minimum E&O inventory value Flex can achieve assuming the company can invest up to B in additional raw materials?

We wish to solve the following Mixed-Integer Linear Program (MILP):

Objective:

We wish to minimize the E&O inventory value after production/depletion:

$$(P_1(B)) \qquad \min_{\mathbf{Inv}_{\cdot,\mathbf{1}},\mathbf{x},\mathbf{y}} \sum_{i \in I^{BW}} C_i \; Inv_{i,1} \tag{21}$$

Constraints:

The inventory after production/depletion is equal to the initial inventory minus production:

$$Inv_{i,1} = Inv_{i,0} + x_i - \sum_{\substack{j \in I^{FG}: \\ \beta_{i,j} > 0}} \beta_{i,j} y_j, \qquad i \in I^{BW}, \tag{22}$$

Sourcing of raw materials (supplies) is bounded above:

$$x_i \leq S_i, \qquad i \in I^{BW}, \tag{23}$$

Production of finished goods (demand) is bounded above:

$$y_i \leq D_i, \qquad i \in I^{FG}, \tag{24}$$

Investments in additional raw materials are bounded above:

$$\sum_{i \in I^{BW}} C_i x_i \leq B. \tag{25}$$

2.3 Problem Formulation #2

We studied a variation of Problem Formulation #1 (which is not identical, but very similar). For this variation, we re-define B as the maximum E&O inventory value. This problem solves the following:

What is the minimum investment in additional raw materials Flex can make in order to lower its E&O inventory value to B or below?

We wish to solve the following MILP:

Objective:

We wish to minimize the investment in additional raw materials:

$$(P_2(B)) \qquad \min_{\mathbf{Inv}_{\cdot,\mathbf{1}},\mathbf{x},\mathbf{y}} \sum_{i \in I^{BW}} C_i \, x_i \tag{26}$$

Constraints:

The inventory after production/depletion is equal to the initial inventory minus production:

$$Inv_{i,1} = Inv_{i,0} + x_i + y_i - g_i - \sum_{\substack{j \in I^{FG} \cup I^{SFG}: \\ \beta_{i,j} > 0}} \beta_{i,j} y_j, \qquad i \in I^{BW}, \tag{27}$$

Sourcing of raw materials (supplies) is bounded above:

$$x_i \leq S_i, \qquad i \in I^{BW}, \tag{28}$$

Production of finished goods (demand) is bounded above:

$$y_i \leq D_i, \qquad i \in I^{FG}, \tag{29}$$

E&O inventory value is bounded below:

$$\sum_{i \in I^{BW}} C_i Inv_{i,1} \leq B. \tag{30}$$